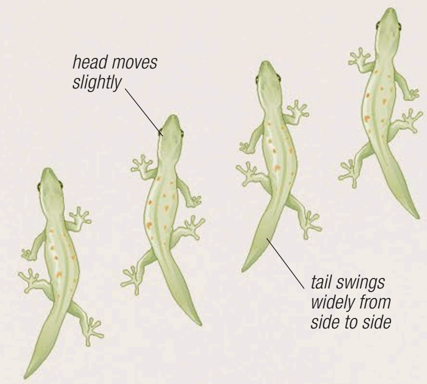
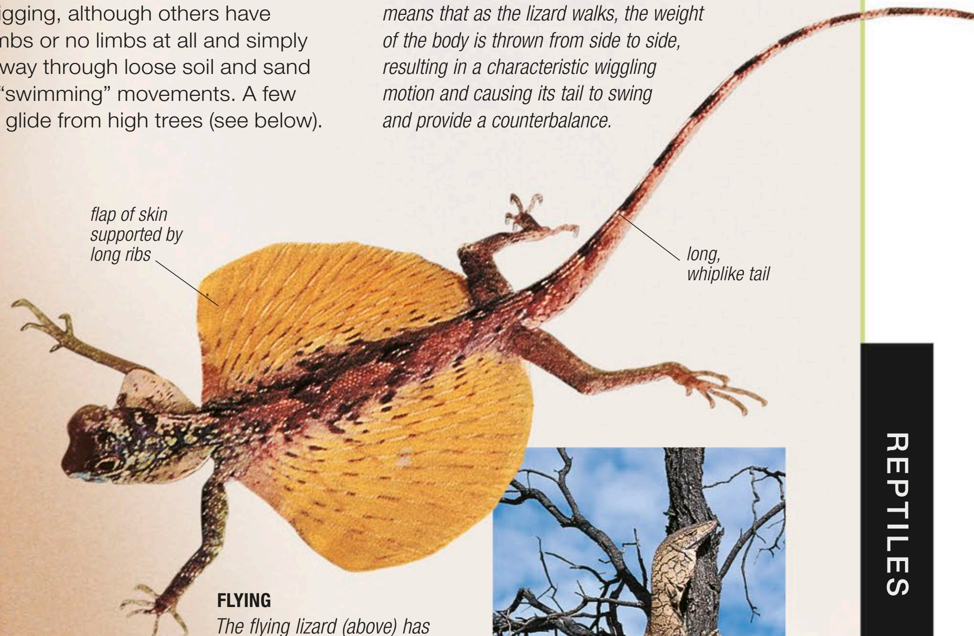
Movement

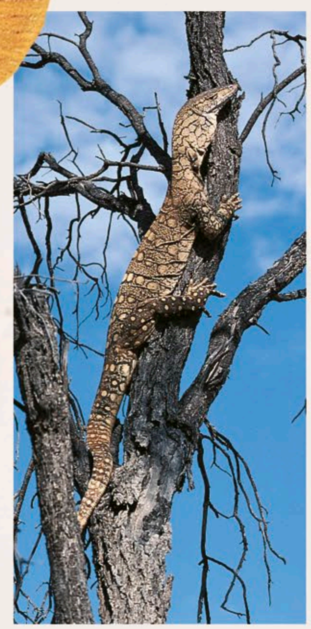
Most lizards move by walking, climbing, or burrowing. When walking, they usually use all 4 limbs (see right), although some species lift their front legs off the ground while running at high speed; basilisk lizards can even run for short distances over the surface of water in this way, helped by broad feet that spread their weight. Some lizards are agile climbers. To improve their grip, they may have long, sharp claws, while some have a prehensile tail. Among the most impressive climbers are geckos, which can cling to smooth, vertical or even overhanging surfaces using adhesive pads on the tips of their digits (see p.417). Burrowing lizards may use powerful front limbs for digging, although others have reduced limbs or no limbs at all and simply force their way through loose soil and sand with rapid "swimming" movements. A few lizards can glide from high trees (see below).



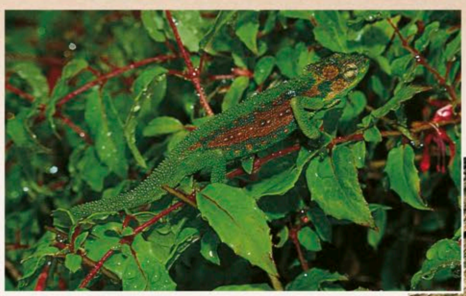
WALKING
A lizard's limbs are held at right angles to the body, rather than supporting it from underneath. This means that as the lizard walks, the weight of the body is thrown from side to side, resulting in a characteristic wiggling motion and causing its tail to swing and provide a counterbalance.



FLYING
The flying lizard (above) has flaps of skin, that act like wings, stretched between elongated ribs. Some other species have similar flaps between their toes. In all cases, the lizards move from tree to tree by gliding rather than powered flight.



CLIMBING TREES
Some large lizards, such as this perentie (right), are surprisingly agile climbers. Perenties usually climb only to escape predators, but other lizards, especially chameleons, spend the greater part of their lives in trees.



CAMOUFLAGE
Chameleons (such as the Cape dwarf chameleon, left) have a remarkable ability to alter the appearance of their skin. Not only does the color change, but markings, such as lines and bars, may appear and disappear.

FIGHTING DRAGONS
The Komodo dragon (right) is the largest of all lizards. When attacking its prey, it usually relies on its jagged teeth and sharp claws. However, when defending itself from another member of its own species, it will often turn its back and thrash its powerful tail.

Defense

Lizards use various active and passive strategies to avoid the many other animals that prey upon them. Other than running away, passive strategies include camouflage and mimicry of inanimate objects, such as sticks and leaves. Some species can alter their coloration, while others make themselves inconspicuous by pressing their bodies close to rocks or tree trunks to avoid casting a shadow. Active strategies include the voluntary loss of the tail (see Regeneration, opposite), and threat displays, which often involve warning coloration (as in blue-tongued skinks) or body enlargement (as in the frilled lizard). Large species, such as monitors, as well as some smaller ones, defend themselves by biting, scratching, or lashing their tail. Two species of lizards, the Gila monster and the Mexican beaded lizard, are venomous.